DO NOW — RECALL

# Lesson 087 · Algorithm Design and Testing Challenge

Name: _____ Class: _____ Date: _____

**1. What is an invariant?**

**2. Why is sorting unnecessary for just finding one maximum?**

**3. What is nullable int?**

**4. Binary search for 8 in [2,4,6,8,10]. Give midpoint indexes and result.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

A condition maintained while processing data.

### 2. Model answer

A single scan can find it in O(n).

### 3. Model answer

An int? that can be an integer or null.

### 4. Model answer

mid2=6, then mid3=8; found index3.

### By the end, you can…

• Specify output when inputs are empty, and preserve correctness on all-negative arrays.

• Maintain invariant: best is the largest value among those inspected so far.

• Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

### Vocabulary — know these terms

maximum — Largest value in set.

invariant — Maintained fact during loop.

nullable — Type allowing absence/null.

complexity — Resource growth as input increases.

NEW LEARNING

# Read, annotate and explain

## Observe

Specify output when inputs are empty, and preserve correctness on all-negative arrays.

## Explain

Maintain invariant: best is the largest value among those inspected so far.

## Evaluate

Compare against alternative sorting-based solution: an O(n) scan avoids O(n log n) sort when only the maximum is needed.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Algorithm Design and Testing Challenge · STAGE 1; Initial best=−5

**New Learning**

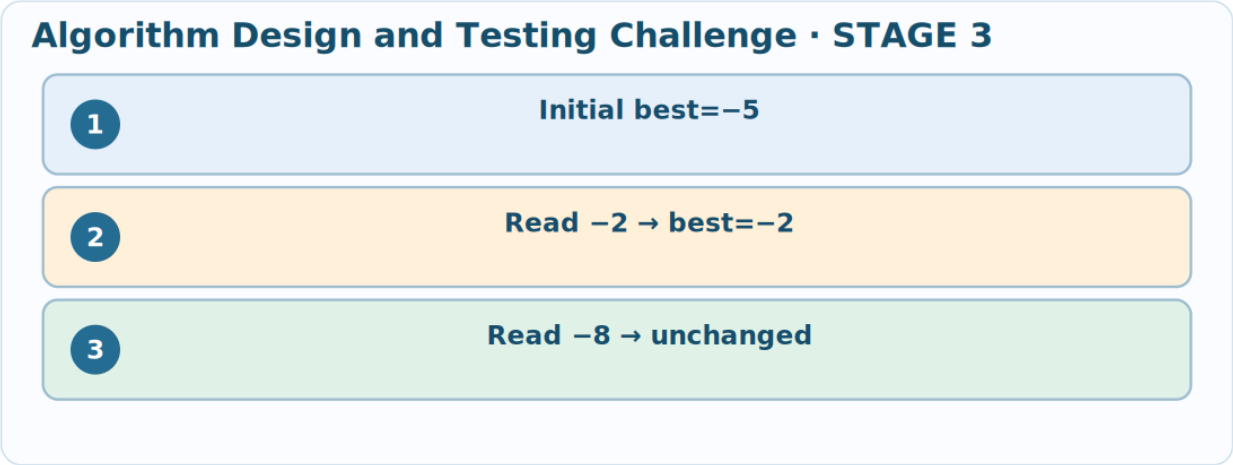



Algorithm Design and Testing Challenge · STAGE 2; Initial best=−5; Read −2 → best=−2

# New Learning · original lesson diagrams

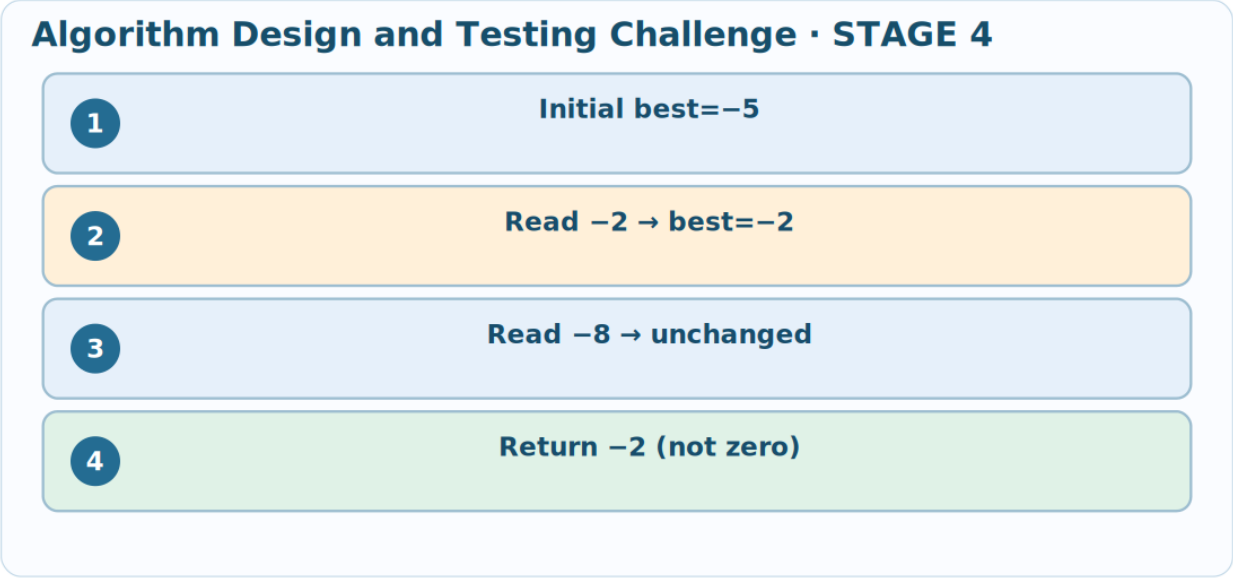
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Algorithm Design and Testing Challenge · STAGE 3; Initial best=−5; Read −2 → best=−2; Read −8 → unchanged

### New Learning




Algorithm Design and Testing Challenge · STAGE 4; Initial best=−5; Read −2 → best=−2; Read −8 → unchanged; Return −2 (not zero)

I DO — WORKED EXAMPLE

# Follow the worked method

### Trace Maximum([7,3,10,4]).

### Model response

The invariant tracks largest seen, not last examined.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Trace Maximum([7,3,10,4]).

The invariant tracks largest seen, not last examined.




IDO · METHOD; Trace Maximum([7,3,10,4]).; best7→7→10→10, returns10.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY

Trace different Maximum([-9,-4,-11]).

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Trace Maximum([-5,-2,-8]) showing all changes and final value. [6]

### Question 2

Write algorithm for maximum of array, including empty input case. [7]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 3

Compare O(n) maximum scan with sorting first and describe useful test coverage. [6]

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**Initialising maximum to zero works for every integer array.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

## GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

## Guided 1 — CHECK

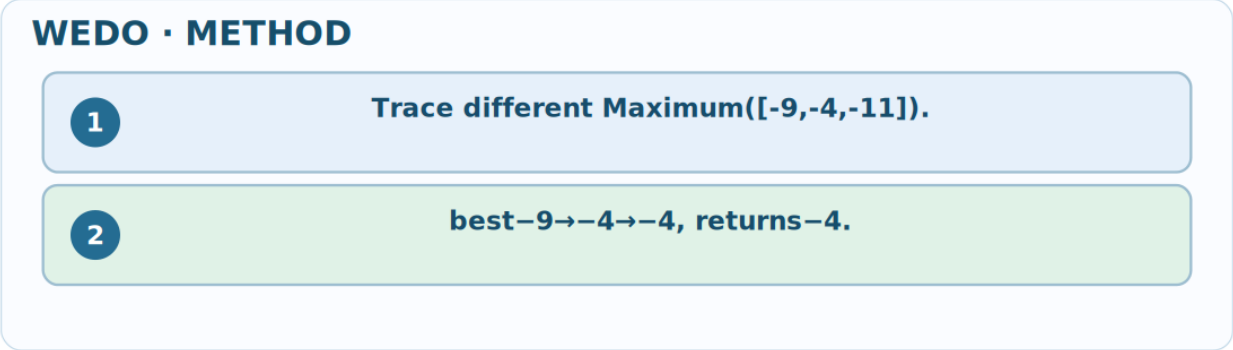
Model response
Starting best at0 would be incorrect for all-negative arrays.

# We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Trace different Maximum([-9,-4,-11]).**

Starting best at0 would be incorrect for all-negative arrays.




WEDO · METHOD; Trace different Maximum([-9,-4,-11]).; best−9→−4→−4, returns−4.